

Blood Cancer Detection with Microscopic Images Using Machine Learning



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Abstract *K*-means transformation, histogram equalization, linear contrast stretching, and share-based features are all used to detect leukemia. A method for automatically classifying leukocytes using microscopic images is proposed. This proposed model used MATLAB to find leukemia cells in healthy blood cells, and it requires no medical equipment or expert and heavily relies on automation. This technology can detect anemia, malaria, vitamin B12 deficiency, and brain tumors. The proposed method correctly identifies WBCs and leukoblasts in images and refines the identification, thresholding, and segmentation phases. This improves WBC counting and overall segmentation accuracy, which leads to better shape feature extraction, which is critical for this problem. New features for this type of analysis must also be studied and analyzed. Finding the most discriminatory features will provide the best accuracy. Determining whether adjacent leukocytes can be separated is critical for counting all leukocytes in an image.

Keywords Leukemia · Leukocytes · WBC · Thresholding · Image processing

1 Introduction

Identifying the leukemia type at an early stage is essential in determining the most appropriate treatment for the specific type of leukemia. It is necessary to perform a

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complete blood count in order to detect leukemia. If the patient's blood cells count is abnormal, it is recommended that they consult with a doctor. As a result, morphological bone marrow and peripheral blood slide analyses are performed to confirm leukemic cells' presence. When a hematologist examines some cells under a light microscope, he will look for abnormalities in the nucleus or cytoplasm of the cells, allowing him to classify the abnormal cells into the various types and subtypes of leukemia present in the sample. It is then up to a hematologist to sort out the abnormal cells and classify them according to the various types and subtypes of leukemia that have been diagnosed within the laboratory. According to this classification, it is possible to predict the clinical behavior of the disease, and treatment should be administered to the patient following the predicted clinical behavior. A person dies when the bone marrow makes too many white blood cells released into the bloodstream. As of right now, there is no known cause for leukemia disease. As a result, the body's immune system gets rid of the wrong white blood cells. Manual detection of leukemia is carried out in pathology, which is both time consuming and expensive due to the high cost of pathology instruments used. As a result, an automatic technique is used to achieve quick and accurate results. The cells are then classified as blasts or normal cells at the end of the section. The rest of the paper is organized as follows: Sect. 2 deals with survey of previous works. The proposed system is explained in Sect. 3, and the results are discussed in Sect. 4. Section 5 concludes the paper.

2 Literature Survey

In order to maximize total rewards, a semi-supervised learning model [1] is used to allow agents to act and interact with their environment while remaining within their learning boundaries. It is common to model real life as a Markov decision process (MDP). Consider the following scenario: A baby is given a television remote control in your home (environment). The first step for an agent is to observe and form a mental picture of the world around him or her (state). The inquisitive baby will then engage in certain activities, such as pressing the remote control (action) and watching how the television responds (next state). Because of the boredom of watching a non-responding television, the baby dislikes it (and thus receives a negative reward), and as a result, the baby will take fewer actions that result in this result (thus updating the policy). As a means of maximizing the total (discounted) rewards, the baby will repeat the process over and over again until he or she discovers a policy (i.e., what to do under different circumstances) that satisfies them. In real-world research, the ultimate goal is to develop a mathematical framework for dealing with real-world problems. When determining the best course of action, there are a variety of methods we can use. These include value-based approaches such as the Q-learning method and policy-based approaches that do not require us to know how effective a particular course of action will be before we take it. The RL method divides leukemia into four

subtypes: ALL, AML, CLL, and CML. The most common type of leukemia is acute lymphoblastic leukemia (ALL).

Using median and unsharp filters, [2] improved the contrast between the cytoplasm, nuclei, and extracellular components by reducing noise in their images. Following the completion of the thresholding process, the Fourier transform and log transform can be used to refine the results even further, if necessary. Following the completion of the preceding steps, a line plot is generated. A pathologist typically uses an optical microscope to diagnose diseases such as AIDS, leukemia, and other blood-related illnesses by identifying white blood cells in the patient's blood (WBCs) [3]. The amount of time, effort, and expense involved in this process necessitate the use of professionals who have a track record of success in similar circumstances. Pathologists can reap significant benefits from the use of a computer-aided diagnosis system because of the advantages it provides. It is the first step in developing an automated diagnosis system that separates white blood cells (WBCs). WBCs are the primary focus of this investigation, distinguishing between these cells and other objects in microscopic images [4]. Three stages are involved: (1) segmentation of white blood cells (WBCs), (2) nuclei extraction from the cell's image, and (3) separation of overlapping cells and nuclei. Thresholding, clustering, and modified watershed algorithms are used in the first stage. They were 92.07% similar when it came to nucleus segmentation, 92.33% similar when it came to cell segmentation, 97.41% similar when it came to precision, and 93.78% similar when it came to sensitivity, according to this method. Precision was also 92.07% the same. According to statistical analysis, this method's results are statistically indistinguishable from the results of manual segmentation [5, 6].

3 Proposed System

It is possible to improve image differentiation by using a picture-preparation system for personal computers, such as histogram equalization, in order to improve image differentiation. This is accomplished through the equitable distribution of power esteems, such as by expanding the power scope of the image appropriately and fairly and equitably. In the bone marrow cells, changes occur that cause the development of leukemia. Patients may succumb to their injuries if treatment is not administered promptly. When you break a bone, you will find a thin substance called bone marrow. This substance is responsible for the formation of the bone, and it can be found inside each one. Red blood cells (RBCs), white blood cells (WBCs), and platelet-rich plasma (PLTs) are the three types of blood cells that can be found in every human body (platelets). The primary focus of this project's attention, which will continue until the end of the year, is the detection of leukemia in children and adults. Because of this, the total number of WBCs is all that we are interested in (leucocytes).

So far, only two types of stem cells have been discovered: embryonic stem cells and adult stem cells. Myeloid cells and lymphoid cells are the two types of cells that exist. An abnormal myeloid blast is formed in the body when a young myeloid stem

cell matures into a mature myeloid cell, a process known as differentiation. This swarm of myeloid cells is responsible for producing an erythrocyte, a white blood cell, and platelets. Lymphoid blasts, which are the only thing that produces white blood cells, can be used to add white blood cells to lymphoid stem cells, allowing them to produce more white blood cells (WBC). The bone marrow is in charge of producing white blood cells harmful to the body, and it does so in large quantities (WBCs). The process of these cells multiplying and growing instead of dying out takes a considerable amount of time. It is the function of abnormal white blood cells to disrupt the normal functions of white blood cells on a daily basis. Various diseases, such as “Leukemia,” have been named after people who have had to deal with the same issues in the past. The body has two types of leukemia: chronic leukemia and acute leukemia, both of which occur as a result of a virus or other infection. Cancer of the bone marrow (CML) is a disease that affects the bone marrow and causes it to malfunction. Normal and abnormal white blood cells perform the same functions when it comes to their actions. As the disease progresses and the number of abnormal white blood cells increases, the situation becomes more serious. Acute leukemia is thought to be caused by abnormal white blood cells, which are more dangerous than normal white blood cells because they do not function similarly to normal white blood cells. In addition, the number of abnormal white blood cells increases as the disease progresses (acute lymphoblastic leukemia).

Figure 1 shows the images of anatomy of human bone. The primary goal of the research was to improve algorithms that can detect disease in human blood images during the early stages of development in order to prevent the disease from progressing further. In the final product, when overlapping cells are separated, it is highly beneficial for the quantitative analysis of images as well as for the recognition of images. If you divide overlapping cells using a watershed algorithm-based algorithm, you will find that the overlapping cells have more favorable findings than when you divide them using a traditional algorithm. As a result of this research, software that can distinguish between overlapping cells in blood images was developed, allowing for more accurate detection of leukemia in the patient population.

Figure 2 shows the typical formation of myeloid cell and lymphoid series of cell. The pipeline followed in myeloid and lymphoid cell is displayed in Fig. 3. Figure 4 shows the methodical steps in identifying WBC.

The process of identifying WBCs is divided into several phases:

- Conversion of the RGB color model to the CMYK color model.
- Equalization of the histogram or stretching of the contrast of the histogram.
- The Zack algorithm is used to segment data based on thresholds.
- The operation of removing the background.

Fig. 1 Bone anatomy

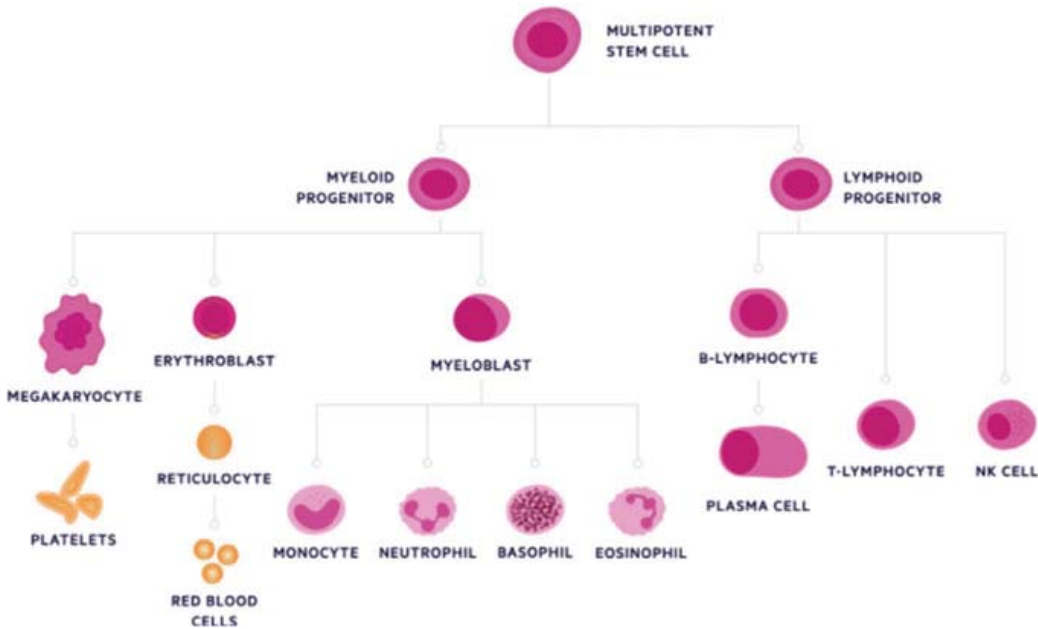
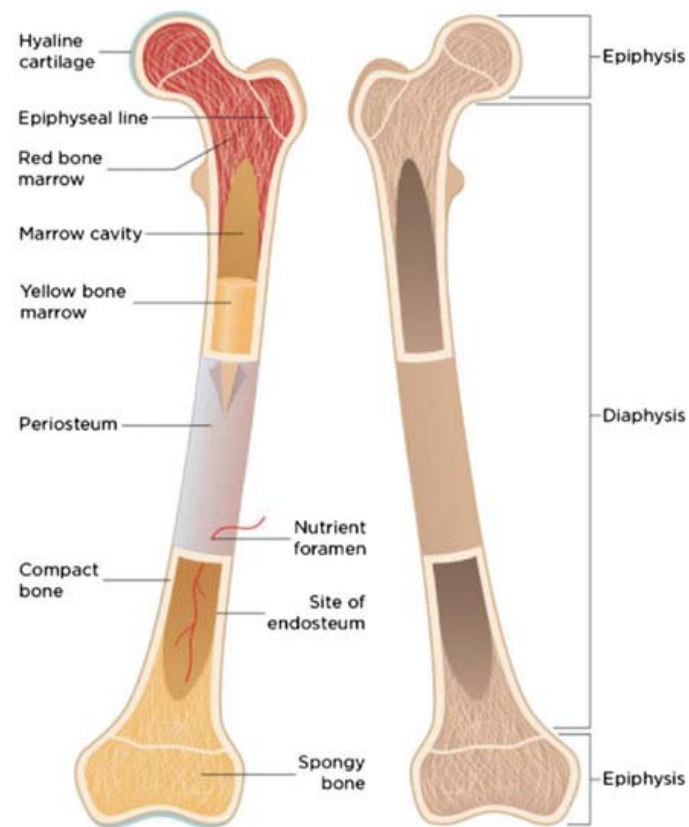


Fig. 2 Formation of myeloid and lymphoid series of cell

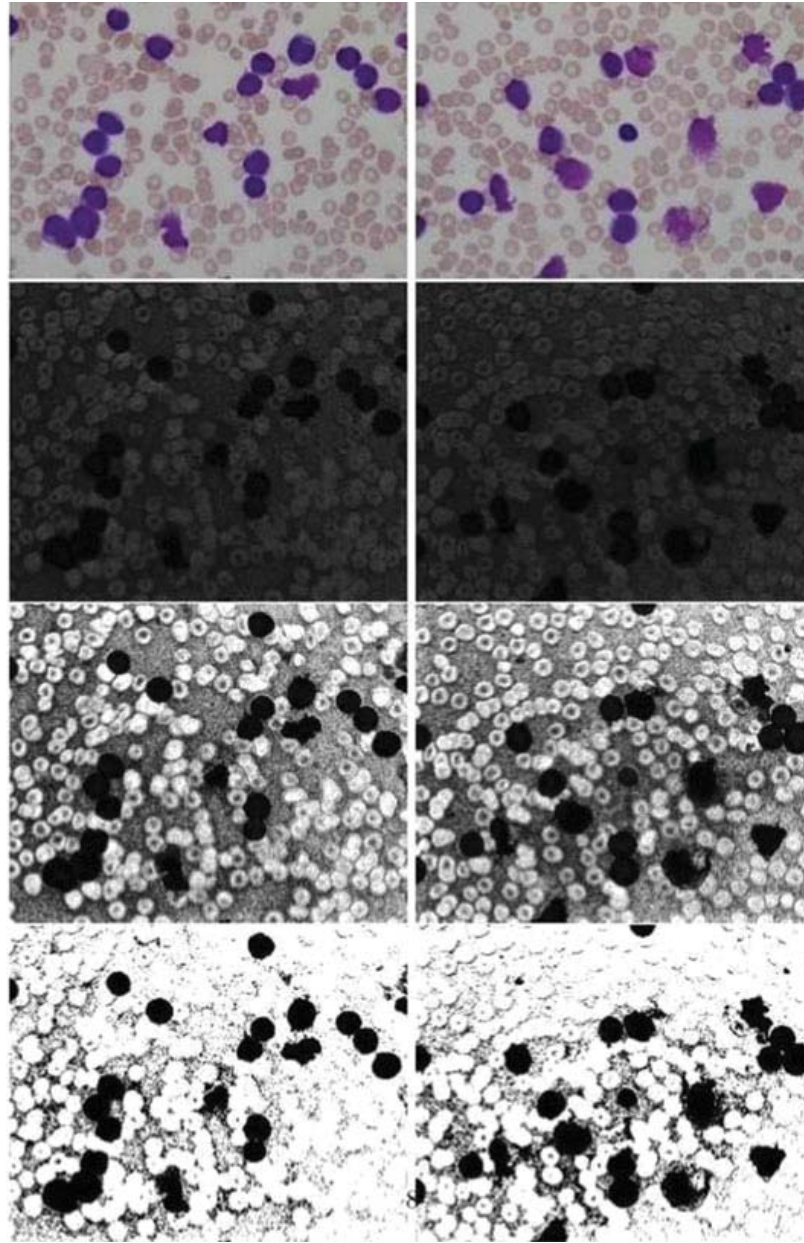


Fig. 3 Pipeline followed

4 Simulation Results

The microscopic input image is depicted in Fig. 5. The proposed method enhances the microscopic image through preprocessing steps, including color conversion, filtering, and histogram equalization.

Following that, the K -means algorithm is used to segment the data. Figure 6 illustrates an enhanced image created using histogram equalization; in this case, the enhancement is done in contrast. The three classes that we have chosen for K -means

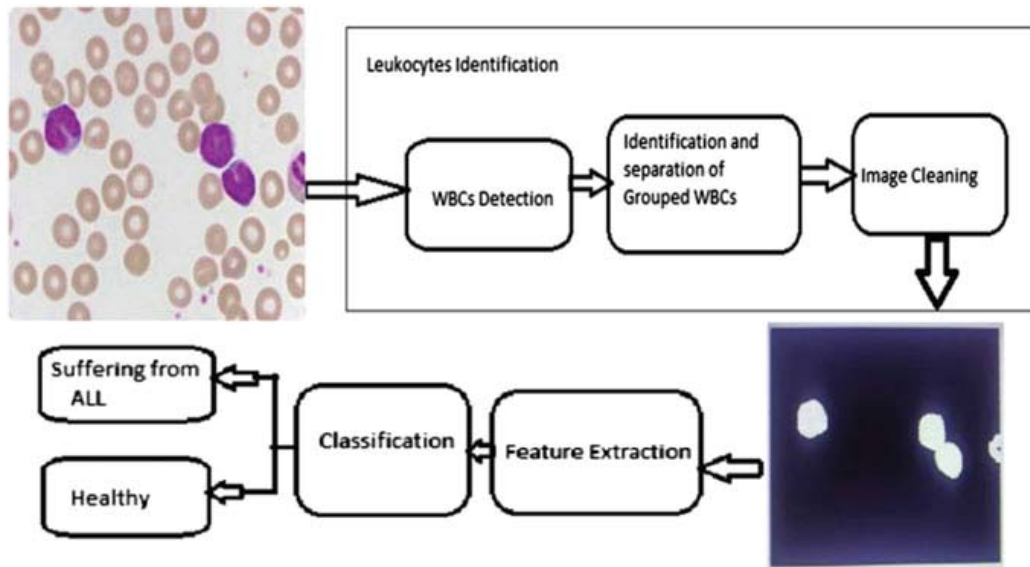
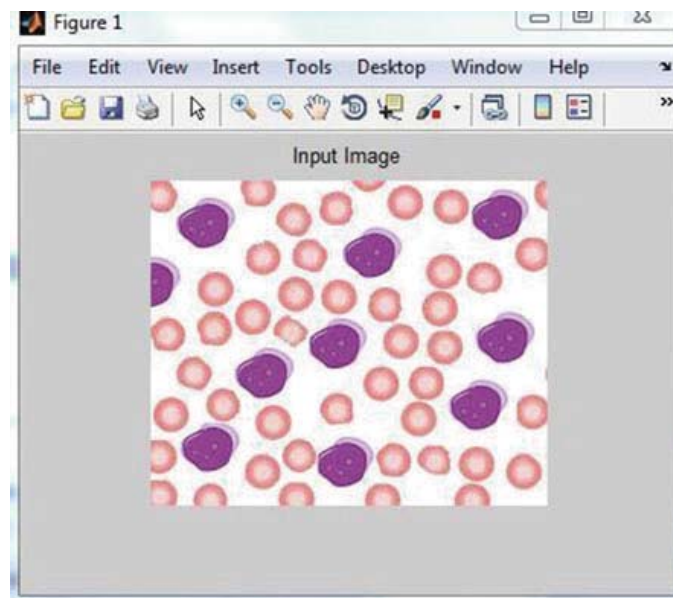


Fig. 4 Steps in WBC identification

Fig. 5 Input image



segmentation are as follows: K is set to 3 in this case because it is necessary to obtain the proper region of interest, which is the nucleus.

Figure 7 shows the output of K -means from the given input image. Figure 8 illustrates the segmented output of K -means which is obtained by the proposed operation.

Compared to the different features' accuracy, the watershed transform's K -means, histogram equalization, linear contrast stretching, and share-based features are all 72.2, 73.7, and 97.8% accurate. The MATLAB software will be used to detect the presence of leukemia cells in healthy individuals' blood cells in this project, which

Fig. 6 Segmented output-1 of K -mean

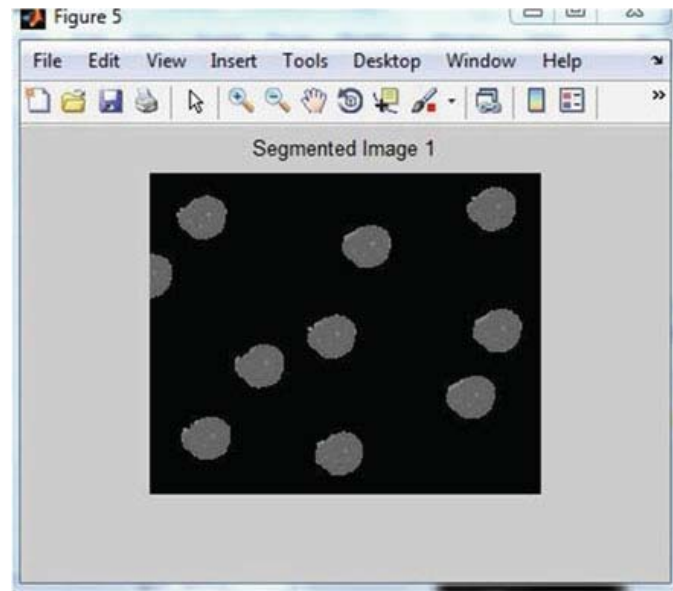
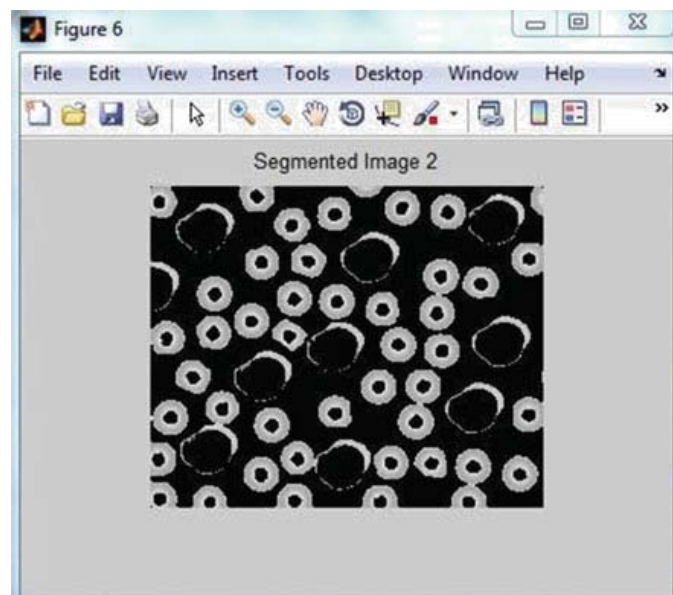
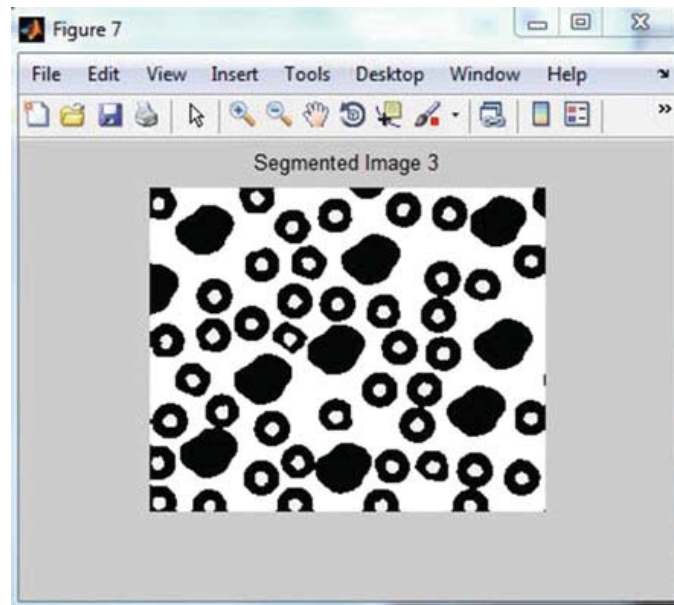


Fig. 7 Segmented output-2 of K -mean



we hope will be completed by the end of the year. The use of an electronic medical device or the presence of a medically trained individual is not required for the implementation of this concept. There is no requirement for large numbers of human resources.

Fig. 8 Segmentation output of *K*-means



5 Conclusion

This technique could be used in the future to diagnose anemia, malaria, vitamin B12 deficiency, and brain tumors, to name a few. An automated procedure is proposed to aid in recognizing acute lymphoblastic leukemia (ALL) using microscopic images, both of which are currently unavailable. Using an image, the proposed method can identify white blood cells (WBCs) and classify leukoblasts with high precision. The previous phases of identification, thresholding, and segmentation will be further developed in this work. White blood cells (WBCs) and overall segmentation accuracy can be improved with a more robust extraction of shape features. Research and analysis of new features will also be necessary for this task. A lot more investigation and analysis are needed. In order to get the most accurate results, it is best to use the most discriminatory features. The proposed method's development may alter the separation of adjacent leukocytes required to account for all leukocytes in an image.

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